Pushforward and Pushback

(%i2) `info:build_info()$info@version;`

(%o2)

5.38.1

(%i2) `reset()$kill(all)$`

(%i1) `derivabbrev:true$`

(%i2) `ratprint:false$`

(%i3) `fpprintprec:5$`

(%i4) `load(linearalgebra)$`

1

Based on Robert Davie Video: The push forward of vectors on manifolds

(%i5) `ζ:[x,y]$`

(%i6) `dim:length(ζ)$`

(%i7) `f:[x^2-y^2,2*x*y]$`

(%i8) `P:[1,1]$`

(%i9) `ldisplay(f_P:at(f,map("=",ζ,P)))$`

$$f_P = [0, 2] \tag{\%t9}$$

(%i10) `J:jacobian(f,ζ);`

$$\begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix} \tag{J}$$

(%i11) `v:[1,0]$`

(%i12) `ldisplay(J_P:at(J,map("=",ζ,P)))$`

$$J_P = \begin{pmatrix} 2 & -2 \\ 2 & 2 \end{pmatrix} \tag{\%t12}$$

(%i13) `ldisplay(f_\*P:list_matrix_entries(J_P.v))$`

$$f_{*P} = [2, 2] \tag{\%t13}$$

2

Based on Robert Davie Video: The Push Forward of Vectors on Manifolds - part 2

(%i14) `ζ:[x,y,z]$`

(%i15) `dim:length(ζ)$`

(%i16) f:[x^2+y*z,y^2+x*z,z^2+x*y]$

(%i17) `P:[1,1,1]$`

(%i18) `ldisplay(f_P:at(f,map("=",ζ,P)))$`

$$f_P = [2,2,2] \tag{\%t18}$$

(%i19) `J:jacobian(f,ζ);`

$$\begin{pmatrix} 2x & z & y \\ z & 2y & x \\ y & x & 2z \end{pmatrix} \tag{J}$$

(%i20) `v:[1,1,0]$`

(%i21) `ldisplay(J_P:at(J,map("=",ζ,P)))$`

$$J_P = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} \tag{\%t21}$$

(%i22) `ldisplay(f_\*P:list_matrix_entries(J_P.v))$`

$$f_{*P} = [3,3,2] \tag{\%t22}$$

3

(%i23) `ζ:[x,y]$`

(%i24) `dim:length(ζ)$`

(%i25) `f:[exp(x)*cos(y),exp(x)*sin(y),x+y]$`

(%i26) `P:[0,0]$`

(%i27) `ldisplay(f_P:at(f,map("=",ζ,P)))$`

$$f_P = [1,0,0] \tag{\%t27}$$

(%i28) `J:jacobian(f,ζ);`

$$\begin{pmatrix} e^x \cos(y) & -e^x \sin(y) \\ e^x \sin(y) & e^x \cos(y) \\ 1 & 1 \end{pmatrix} \tag{J}$$

(%i29) `v:[1,-1]$`

(%i30) `ldisplay(J_P:at(J,map("=",ζ,P)))$`

$$J_P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \tag{\%t30}$$

(%i31) `ldisplay(f_\*P:list_matrix_entries(J_P.v))$`

$$f_{*P} = [1, -1, 0] \tag{\%t31}$$

4

Based on Robert Davie Video: The Pushforward of Vectors on Manifolds - 3

(%i32) `ξ:[ct,r,θ,ϕ]$`

(%i33) `dim:length(ξ)$`

(%i34) `depends(r\*,r)$`

(%i35) `f:[ct+r\*,r,θ,ϕ]$`

(%i36) `J:jacobian(f,ξ);`

$$\begin{pmatrix} 1 & r*_r & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{J}$$

(%i37) `gradef(r\*,r,1/(1-2*G*M/c^2/r))$`

(%i38) `J:jacobian(f,ξ);`

$$\begin{pmatrix} 1 & \frac{1}{1-\frac{2GM}{c^2 r}} & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \tag{J}$$

(%i39) `v_t:[1,0,0,0]$`

(%i40) `ldisplay(f\*v_t:list_matrix_entries(J.v_t))$`

$$f * v_t = [1, 0, 0, 0] \tag{\%t40}$$

(%i41) `v_r:[0,1,0,0]$`

(%i42) `ldisplay(f\*v_r:list_matrix_entries(J.v_r))$`

$$f * v_r = [\frac{1}{1 - \frac{2GM}{c^2 r}}, 1, 0, 0] \tag{\%t42}$$

```
(%i43) v_θ:[0,0,1,0]$
(%i44) ldisplay(f\*v_θ:list_matrix_entries(J.v_θ))$
```

$$f * v_\theta = [0, 0, 1, 0] \quad (\%t44)$$

```
(%i45) v_ϕ:[0,0,0,1]$
(%i46) ldisplay(f\*v_ϕ:list_matrix_entries(J.v_ϕ))$
```

$$f * v_\phi = [0, 0, 0, 1] \quad (\%t46)$$

5

Based on Robert Davie Video: The Push Forward of Vectors on Manifolds - part 4

$$(\mathrm{g} \circ \mathrm{f})(p) = \mathrm{g}\left(\mathrm{f}(p)\right) \in \mathbb{R}$$

$$\mathrm{f}_{*\,p}(v)(\mathrm{g}) = v\left(\mathrm{g} \circ \mathrm{f}\right)$$

```
(%i47) kill(labels)$
(%i1) ζ:[x,y]$
(%i2) dim:length(ζ)$
(%i3) f:[u,w]:[x^2-y^2,2*x*y]$
```

Calculate $\mathrm{g} \circ \mathrm{f}$

```
(%i4) g:u+w;
```

$$-y^2 + 2xy + x^2 \quad (\mathrm{g})$$

```
(%i5) P:[1,1]$
(%i6) v:[1,0]$
(%i7) dg:makelist(diff(g,k),k,ζ);
```

$$[2y + 2x, 2x - 2y] \quad (\mathrm{dg})$$

```
(%i8) at(dg,map("=",ζ,P));
```

$$[4, 0] \quad (\%o8)$$

6

Based on Robert Davie Video: The Pullback of 1-forms

$$(\mathrm{f}^*\omega)_p(v_p) = \omega_{\mathrm{f}(p)}(\mathrm{f}_* v_p) \in \mathbb{R}$$

$$\mathrm{f}^*\omega = \omega_j\,(\mathrm{f}(p))\,\frac{\partial \mathrm{f}^j}{\partial x^i}\mathrm{d}x^i$$

(%i9) `if get('cartan,'version)=false then load(cartan)$`

(%i10) `ζ:[x,y]$`

(%i11) `dim:length(ζ)$`

(%i12) `f:[x^2-y^2,2*x*y]$`

(%i13) `J:jacobian(f,ζ);`

$$\begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix} \qquad \text{(J)}$$

(%i14) `ξ:[z_1,z_2]$`

(%i15) `map("=",ξ,f);`

$$[z_1 = x^2 - y^2, z_2 = 2xy] \qquad (\%o15)$$

(%i16) `ldisplay(dξ:makelist(concat(d,k),k,ξ))$`

$$d\xi = [dz_1, dz_2] \qquad (\%t16)$$

(%i17) `init_cartan(ζ)$`

(%i18) `ldisplay(cartan_basis,cartan_coords,cartan_dim,extdim)$`

$$cartan\_basis = [dx, dy] \qquad (\%t18)$$

$$cartan\_coords = [x, y] \qquad (\%t19)$$

$$cartan\_dim = 2 \qquad (\%t20)$$

$$extdim = 2 \qquad (\%t21)$$

(%i22) `list_matrix_entries(J.cartan_basis);`

$$[2x\;dx - 2y\;dy, 2x\;dy + 2y\;dx] \qquad (\%o22)$$

(%i23) `ldisplay(α:ev(z_2,map("=",ξ,f)))$`

$$\alpha = 2xy \qquad (\%t23)$$

(%i24) `makelist(diff(α,k),k,ζ);`

$$[2y, 2x] \qquad (\%o24)$$

(%i25) `ldisplay(f\*ω=%.cartan_basis)$`

$$f \cdot \omega = 2x\;dy + 2y\;dx \qquad (\%t25)$$

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Based on Robert Davie Video: The Pullback of 1-forms - part 2

Example 1

(%i26) `kill(x,y,f,J)$`

(%i27) `ζ:[x,y]$`

(%i28) `dim:length(ζ)$`

(%i29) `f:[x^2-y^2,2*x*y]$`

(%i30) `J:jacobian(f,ζ);`

$$\begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix} \tag{J}$$

(%i31) `init_cartan(ζ)$`

(%i32) `ldisplay(ω:c*dz_1+b*dz_2)$`

$$\omega = b\, dz_2 + c\, dz_1 \tag{\%t32}$$

(%i33) `makelist(coeff(ω,k),k,dξ);`

$$[c, b] \tag{\%o33}$$

(%i34) `ldisplay(f\*=edit(transpose(%).J.cartan_basis))$`

$$f\cdot = (2bx - 2cy)\ dy + (2by + 2cx)\ dx \tag{\%t34}$$

Example 2

(%i35) `kill(x,y,f,J)$`

(%i36) `ζ:[x,y]$`

(%i37) `dim:length(ζ)$`

(%i38) `f:[x+y,x-y]$`

(%i39) `J:jacobian(f,ζ);`

$$\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \tag{J}$$

(%i40) `ldisplay(ω:z_2*dz_1+z_1*dz_2)$`

$$\omega = dz_1 z_2 + dz_2 z_1 \tag{\%t40}$$

(%i41) `makelist(coeff(ω,k),k,dξ);`

$$[z_2, z_1] \tag{\%o41}$$

(%i42) `ldisplay(f\*ω:edit(transpose(%).J.cartan_basis))$`

$$f \cdot \omega = (z_2 - z_1)\, dy + (z_2 + z_1)\, dx \tag{\%t42}$$

(%i43) `ldisplay(f\*ω:subst(map("=",ξ,f),f\*ω))$`

$$f \cdot \omega = 2x\, dx - 2y\, dy \tag{\%t43}$$

8

Based on Robert Davie Video: The Pullback of 2-forms

For any point $p \in M$ **and vectors** $v, w \in T_pM$:

$$(\mathrm{f}^*\omega)_p\,(v,w) = \omega\,(\mathrm{f}(p))\,(D_p\mathrm{f}(v), D_p\mathrm{f}(w))$$

(%i44) `kill(x,y,f,J)$`

(%i45) `ζ:[x,y]$`

(%i46) `dim:length(ζ)$`

(%i47) `f:[x`^{2}`-y`^{2}`,2*x*y]$`

(%i48) `J:jacobian(f,ζ);`

$$\begin{pmatrix} 2x & -2y \\ 2y & 2x \end{pmatrix} \tag{J}$$

(%i49) `factor(determinant(J));`

$$4\left(y^2 + x^2\right) \tag{\%o49}$$

(%i50) `Jdet:`$\surd$`(factor(determinant(transpose(J).J)));`

$$4\left(y^2 + x^2\right) \tag{Jdet}$$

(%i51) `init_cartan(ζ)$`

(%i52) `ldisplay(df:ext_diff(f))$`

$$df = [2x\, dx - 2y\, dy, 2x\, dy + 2y\, dx] \tag{\%t52}$$

(%i53) `ldisplay(f\*ω:factor(edit(df[1] ~ df[2])))$`

$$f \cdot \omega = 4\left(y^2 + x^2\right)\ dx\, dy \tag{\%t53}$$